

CHAPTER 1

Working with MS Access



Database

Tables

Queries

Forms & Reports

What is a Database?



The Analogy — School Library

Every book has a card:

Title / Author / Shelf No.

All cards organised together

= your DATABASE!

Add all the cards for all the books and you have a library catalogue — structured, searchable, instantly findable.

Definition

A database is a collection of data or information of a similar set of records, stored in an organised way.

Real-life examples:



Bank accounts



Flight schedules



Hospital records



Student grades

What is a DBMS?

The Librarian Analogy

Library = Database

Librarian = DBMS

The librarian FINDS your book, BLOCKS
unauthorised people, BACKS UP records,
and lets MANY students use it at once!



Security

Controls who can see or change
data



Backup & Recovery

Saves & restores data if
something goes wrong



Concurrency

Multiple users can access data
simultaneously



Data Integrity

Rules ensure data stays
accurate and consistent

Popular DBMS Software:

Oracle

MS-Access

SQL Server

Clipper

Foxpro

★ We study MS Access in this chapter

MS Access — What Can It Do?



Organise data into manageable related units



Enter, modify and locate data quickly



Extract subsets of data based on specific criteria



Create custom forms and reports



Automate common database tasks



Graph database relationships visually



Create macros to automate various functions



Build complete database applications

4 Key Components of MS Access



Tables

Store raw data

≈ Filing Cabinet

Rows = Records

Columns = Fields

Each table = one topic



Queries

Search & filter

≈ Search Clerk

Find records by conditions

Sort and extract data

Answer questions fast



Forms

Easy data entry

≈ Admission Form

User-friendly screens

One record at a time

Prevents entry errors



Reports

Print & present

≈ Marksheet

Formatted printable output

Summarise & group data

Share results neatly

Data Types — The Right Container for Every Value

Analogy: You wouldn't store soup in a paper bag! Match the container to what's inside — just like data types match the kind of value.



Short Text

Storage: Up to 255 chars

eg. Priya, Class 7A

Names, codes, labels



Number

Storage: 1 to 8 bytes

eg. 85, 42, 1000

Marks, age, quantity



Date / Time

Storage: 8 bytes

eg. 15/08/2024

Birthdays, deadlines



Yes / No

Storage: 1 bit

eg. Yes or No

Pass/Fail, Active/Not



Long Text

Storage: Up to 1 GB

eg. Detailed remarks

Long descriptions



AutoNumber

Storage: 4 bytes

eg. 1, 2, 3, 4...

Auto unique IDs

Creating a New Database in MS Access

1



Open MS Access 2016

Start → All Programs → Microsoft Office 2016 → Access (or double-click desktop shortcut)

2



Choose Blank Desktop Database

Click 'Blank Desktop Database' on the welcome screen

3



Name Your Database & Choose Location

Type a file name → click Browse to choose folder → click OK

4



Click Create!

Access creates the file and opens a blank Table1 ready for you

5



Blank Table Ready — Start Adding Data

ID column is added automatically. Click 'Click to Add' to add more columns.

Two Ways to Create a Table

Datasheet View

Quick Start

- ✓ Type data straight into a grid — like Excel
- ✓ Good for quick, simple tables
- ✓ ID field added automatically
- ✓ Less control over field settings

Analogy: Jump into writing in your notebook without setting up headings first — fast but less organised.

VS

Design View ★ Recommended

Most professional

- ✓ Plan field names and data types first
- ✓ Full control over every field
- ✓ Set primary key, rules, constraints
- ✓ Preferred method in real workplaces

Analogy: Read the full blueprint before building the desk — precise and professional.

Working with Records — Add, Edit, Delete



Add New Record

- 1 Switch to Datasheet View
- 2 Click empty row at bottom
- 3 Type data in each field
- 4 Press Tab to move to next field
- 5 Access saves automatically



Edit a Record

- 1 Click cell you want to change
- 2 Pencil icon appears — edit mode
- 3 Type the new value
- 4 Press Tab or click away to save
- 5 Can also Copy & Paste records



Delete a Record

- 1 Select record (click row arrow)
- 2 Right-click → Delete Record
- 3 Access shows a WARNING dialog
- 4 Click YES to permanently delete
- 5 Cannot be undone — be careful!

The MS Access 2016 Interface

Ribbon — All Commands

HOME CREATE EXTERNAL DATA DATABASE TOOLS

View

Paste

Cut

Copy

Filter

Find

Select

New

Save

Delete

All Access Objects

► Tables

► Queries

► Forms

► Reports

ID

Student Name

Class

Section

Age

Address

1

Priya

7

A

12

Hyderabad

2

Aryan

7

B

12

Chennai

3

Sana

7

A

13

Bangalore

Record: 1 of 3



No Filter Search

Navigation
Pane

Status Bar (record navigation)

Backstage View & Quick Access Toolbar

Backstage View (File Tab)

Like opening the car bonnet — for big file tasks:

1. Click the FILE tab on the Ribbon
2. Backstage View opens full screen
3. Options: New / Open / Save / Info
4. To open existing DB → Open → Browse
5. .accdb is the standard Access 2016 file format

Quick Access Toolbar (QAT)

Your shortcut strip — always visible top-left:



Save

Save database — shortcut: Ctrl+S



Undo

Reverse last action — Ctrl+Z



Redo

Redo what you undid — Ctrl+Y

Tip: Right-click any Ribbon button → 'Add to Quick Access Toolbar' to customise your shortcuts!

Chapter 1 — Key Takeaways



Database

Organised collection of similar records — like a digital filing cabinet



DBMS

Software that stores, finds, secures and backs up your database



Tables

Building blocks — rows are records, columns are fields



Queries

Search and filter data based on conditions you set



Forms & Reports

Forms for data entry; Reports for neat printed output



Data Types

Each field must have the right type: Text, Number, Date, Yes/No...

Next Class: Open MS Access and build your own Student Database from scratch!